

Project Title:- File Integrity Checker

Objective:-

To detect unauthorized changes in files using SHA-256 hashing and maintain data integrity. This project helps users verify whether a file has been altered after its original hash value was generated.

Problem Statement:-

Important files can be modified by malware, viruses, or unauthorized users without the owner's knowledge. Such modifications can compromise data security and reliability. This project helps detect these changes by comparing SHA-256 hash values of files.

Research Findings:-

SHA-256 generates a unique hash value for every file.

Even a one-character change in a file creates a completely different hash.

Hashing is widely used in cybersecurity for integrity verification.

SHA-256 is considered secure and is commonly used in security applications.

Hash values help identify unauthorized modifications in files.

Methodology:-

Create a sample file for testing.

Read the contents of the file.

Generate a SHA-256 hash value using Python.

Display the generated hash value.

Compare hash values whenever the file is checked again.

If the hash changes, it indicates that the file has been modified.

Tools Used:-

Python

PyDroid 3

hashlib Library

GitHub

Android Device

Implementation:-

```
import hashlib
```

```
with open("test.txt", "w") as f:  
    f.write("Hello World")
```

```
with open("test.txt", "rb") as f:  
    data = f.read()
```

```
hash_value = hashlib.sha256(data).hexdigest()
```

```
print("File Created Successfully!")  
print("SHA-256 Hash:")  
print(hash_value)
```

Results:-

Successfully created a sample file automatically.
Generated a SHA-256 hash value for the file.
Verified file content using cryptographic hashing.
Demonstrated how file integrity can be checked.
Produced accurate and consistent hash results.

Advantages:-

Easy to use and implement.
Detects unauthorized file modifications.
Improves data security and reliability.
Fast and efficient integrity verification.
Useful for cybersecurity applications.

Future Scope:-

Real-time file monitoring.
Automatic integrity checking for multiple files.
Email or notification alerts on file modification.
Graphical User Interface (GUI) support.
Cloud-based file integrity monitoring.

Screenshot:-

2:29



TAB



File Created Successfully!

SHA-256 Hash:

a591a6d40bf420404a011733cfb7b190d62c65bf0bcda32b57b277d9ad9f146e

[Program finished]



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Conclusion:-

The File Integrity Checker is a simple yet effective cybersecurity project that uses SHA-256 hashing to verify file integrity. It helps users detect unauthorized modifications, protect important data, and maintain file security. The project demonstrates the practical use of cryptographic hashing in real-world cybersecurity applications.